

LESSON 149b (0.5 GROUND, 1.5 DUAL, and 0.3 INSTRUMENT)

Lesson Objective

During this lesson, the applicant must be able to perform each of the areas of operation in the current FAA Private Pilot ACS.

Pre-Brief

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Instructor Guidance

- This is the mock EOC practical flight — conduct it as close to the actual End-of-Course test experience as possible. The student should be treated as if this is the real test, including the go/no-go decision, plan of action, and all required maneuvers being evaluated to ACS standards without coaching.
- All maneuvers are "student says, student does" with no procedural assistance. Intervene only for safety. Your silence during the flight is intentional — the student must demonstrate they can operate independently, which is the entire purpose of this lesson.
- Ensure pre-maneuver checks are completed before every maneuver without prompting. Skipping or rushing clearing turns is one of the most common deficiencies noted on practical tests and must be identified now if it is a habit.
- Simulate the diversion and lost procedures scenarios during the cross-country portion. The student must work through both without assistance — their ability to maintain aircraft control while managing navigation and decision-making simultaneously is a direct indicator of EOC readiness.
- Do not advance the student to the actual EOC in Lesson 150 if performance falls meaningfully short of ACS standards in any area. An honest assessment now prevents a failed EOC and protects the student's record.
- Use the post-brief to give a complete debrief covering every maneuver, every procedural item, and every decision made during the flight. The student should leave this lesson knowing exactly where they stand and what, if anything, still needs improvement before the EOC.

Lesson Content

Checklist Procedures

- Have the student run through all checklists.
- Incorporate flows for memory items.
 - Student should be able to perform all briefings.

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

- Walk around the airplane and quiz the student on:
 - Antennas.
 - Aircraft structure.
 - Flight controls.
 - Systems.
 - Placards and markings.

Operation of the Communications and Navigation Units

- Ensure that the student can proficiently use COMS and NAV units.

Engine Starting

- Verify correct procedures.

Radio Communication Procedures and ATC Phraseology

- The student should be able to conduct radio calls at towered and non-towered airports.
- The student should be able to conduct radio calls in the practice area.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!
- Ensure correct right-of-way rules are being used.

Taxiing

- Verify correct procedures for the taxi check.
- Taxi speeds should be at a brisk walking pace (5kts GS).
- Power to idle before braking.
 - No riding the breaks.
 - Taxi instructions must be briefed.

Normal and crosswind takeoff

- Student says, student does.
 - Verify correct procedure.
 - Ensure proper crosswind correction.
- Standards:
 - Rotate and lift off at the recommended airspeed and accelerate to V_Y .
 - Establish a pitch attitude to maintain the manufacturer's recommended speed or V_Y , +10/-5 knots.
 - Maintain V_Y +10/-5 knots to a safe maneuvering altitude.
 - Maintain directional control and proper wind-drift correction throughout takeoff and climb.

Short-field takeoff and maximum performance climb

- Student does, student says:

- Verify correct procedure.
- Ensure that the student holds the brakes until full static RPM.
- Verify that the climb out is conducted at 63 knots.
- Verify that the student says "obstacle cleared" if there is an obstacle or simulated obstacle.
- Verify that the student says positive rate of climb before retracting the flaps.
- Standards:
 - Rotate and lift off at the recommended airspeed and accelerate to the recommended obstacle clearance airspeed or VX, +10/-5 knots.
 - Establish a pitch attitude that will maintain the recommended obstacle clearance airspeed or VX, +10/-5 knots until clearing the obstacle or until the airplane is 50 feet above the surface.

Soft-field takeoff and climb

- Student does, student says:
 - Verify correct procedure.
 - Verify that the climb out is conducted at 63 or 79 knots depending on obstacles.
 - Verify that the student says "obstacle cleared" if there is an obstacle or simulated obstacle.
 - Verify that the student says positive rate of climb before retracting the flaps.
- Standards:
 - Establish and maintain a pitch attitude that will transfer the weight of the airplane from the wheels to the wings as rapidly as possible.
 - Lift off at the lowest possible airspeed and remain in ground effect while accelerating to VX or VY, as appropriate.
 - Establish a pitch attitude for VX or VY, as appropriate, and maintain selected airspeed +10/-5 knots during the climb.

Steep Turns

- Student does, student says:
 - Verify correct procedure.
 - Ensure that the student is looking both outside and inside.
 - Verify the roll out happens at ½ the bank angle before the desired heading.
- Standards:
 - Roll into a coordinated 360° steep turn with approximately a 45° bank.
 - Perform the Task in the opposite direction, as specified by evaluator.
 - Maintain the entry altitude ±100 feet, airspeed ±10 knots, bank ±5°, and roll out on the entry heading ±10°.

Maneuvering During Slow Flight

- Student says, student does.
 - Verify correct procedure.
 - Ensure the student is pitching for airspeed and using power to control altitude.

- Standards:
 - Select an entry altitude that will allow the Task to be completed no lower than 1,500 feet AGL (ASEL, ASES).
 - Establish and maintain an airspeed at which any further increase in angle of attack, increase in load factor, or reduction in power, would result in a stall warning (e.g., airplane buffet, stall horn, etc.).
 - Accomplish coordinated straight-and-level flight, turns, climbs, and descents with the airplane configured as specified by the evaluator without a stall warning (e.g., airplane buffet, stall horn, etc.).
 - Maintain the specified altitude, ± 100 feet; specified heading, $\pm 10^\circ$; airspeed, $+10/-0$ knots; and specified angle of bank, $\pm 10^\circ$.

Power-off stalls

- Student says, student does.
 - Verify correct procedure.
 - Ensure proper coordination.
- Standards:
 - Select an entry altitude that will allow the Task to be completed no lower than 1,500 feet AGL (ASEL, ASES).
 - Configure the airplane in the approach or landing configuration, as specified by the evaluator, and maintain coordinated flight throughout the maneuver.
 - Establish a stabilized descent.
 - Transition smoothly from the approach or landing attitude to a pitch attitude that will induce a stall.
 - Maintain a specified heading $\pm 10^\circ$ if in straight flight; maintain a specified angle of bank not to exceed 20° , $\pm 10^\circ$ if in turning flight, while inducing the stall.

Power-on stalls

- Student says, student does.
 - Verify correct procedure.
 - Ensure proper coordination.
- Standards:
 - Select an entry altitude that will allow the Task to be completed no lower than 1,500 feet AGL (ASEL, ASES).
 - Establish the takeoff, departure, or cruise configuration, as specified by the evaluator, and maintain coordinated flight throughout the maneuver.
 - Set power (as assigned by the evaluator) to no less than 65 percent power.
 - Transition smoothly from the takeoff or departure attitude to the pitch attitude that will induce a stall.
 - Maintain a specified heading $\pm 10^\circ$ if in straight flight; maintain a specified angle of bank not to exceed 20° , $\pm 10^\circ$ if in turning flight, while inducing the stall.
 - Spin awareness:
 - Coordination!

Basic Instrument Maneuvers

- The student must be able to perform the following:
 - Straight and level flight.
 - Standard rate turns and medium level turns.
 - Constant airspeed climbs and descents.
 - Recoveries from unusual attitudes (ensure that the student recovers using the correct procedure depending on nose low or nose high).
- Standards:
 - Maintain altitude ± 200 feet, heading $\pm 20^\circ$, and airspeed ± 10 knots.
 - Level off at the assigned altitude and maintain altitude ± 200 feet, heading $\pm 20^\circ$, and airspeed ± 10 knots.
 - Turn to headings, maintain altitude ± 200 feet, maintain a standard rate turn, roll out on the assigned heading $\pm 10^\circ$, and maintain airspeed ± 10 knots.
 - Recognize unusual flight attitudes; perform the correct, coordinated, and smooth flight control application to resolve unusual pitch and bank attitudes while staying within the airplane's limitations and flight parameters.

Emergency Descent

- Student says, student does.
 - Verify correct procedure.
- Standards:
 - Maintain orientation, divide attention appropriately, and plan and execute a smooth recovery.
 - Use a bank angle between 30° and 45° to maintain positive load factors during the descent.
 - Maintain appropriate airspeed $+0/-10$ knots, and level off at a specified altitude ± 100 feet.

Emergency approach and landing

- Student says, student does.
 - Verify correct procedure.
 - Evaluate choice of landing point.
 - Landing must be made with a head wind.
 - Ensure that the student verifies with the checklists.
- Standards:
 - Establish and maintain the recommended best glide airspeed, ± 10 knots.
 - Select a suitable landing area considering altitude, wind, terrain, obstructions, and available glide distance.
 - Plan and follow a flightpath to the selected landing area considering altitude, wind, terrain, and obstructions.
 - Complete the appropriate checklist.

Systems and Equipment Malfunctions

- Give the student a scenario such as:
 - Low oil pressure, high oil temperature, carburetor icing, lost comms, etc..
 - Ensure that the student can troubleshoot and use the emergency checklist.

Ground Reference Maneuvers

- Rectangular course:
 - Student says, student does.
 - Verify correct procedure.
 - Verify proper wind correction on the various pattern legs.
 - Standards:
 - Enter a left or right pattern, 600 to 1,000 feet above ground level (AGL) at an appropriate distance from the selected reference area, 45° to the downwind leg.
 - Apply adequate wind-drift correction during straight and turning flight to maintain a constant ground track around a rectangular reference area, or to maintain a constant radius turn on each side of a selected reference line or point.
 - Maintain altitude ± 100 feet; maintain airspeed ± 10 knots.
- S-turns:
 - Student says, student does.
 - Verify correct procedure.
 - Both turns must have the same radius.
 - Bank must be adjusted for ground speed/winds.
 - Standards:
 - Enter perpendicular to the selected reference line, 600 to 1,000 feet AGL at an appropriate distance from the selected reference area.
 - Apply adequate wind-drift correction during straight and turning flight to maintain a constant ground track around a rectangular reference area, or to maintain a constant radius turn on each side of a selected reference line or point.
 - Maintain altitude ± 100 feet; maintain airspeed ± 10 knots.
- Turns around a point:
 - Student says, student does.
 - Verify correct procedure.
 - Radius must be the same throughout the turn.
 - Bank must be adjusted for ground speed/winds.
 - Standards:
 - Enter at an appropriate distance from the reference point, 600 to 1,000 feet AGL at an appropriate distance from the selected reference area.
 - Apply adequate wind-drift correction during straight and turning flight to maintain a constant ground track around a rectangular reference area, or to maintain a constant radius turn on each side of a selected reference line or point.
 - Maintain altitude ± 100 feet; maintain airspeed ± 10 knots.

Traffic Patterns

- Student should be able to fly traffic patterns and demonstrate understanding of;
 - Towered and nontowered airport operations.
 - Traffic pattern selection for the current conditions.
 - Right-of-way rules.
 - Use of automated weather and airport information.
- Standards:
 - Identify and interpret airport/seaplane base runways, taxiways, markings, signs, and lighting.
 - Comply with recommended traffic pattern procedures.
 - Correct for wind drift to maintain the proper ground track.
 - Maintain orientation with the runway/landing area in use.
 - Maintain traffic pattern altitude, ± 100 feet, and the appropriate airspeed, ± 10 knots.
 - Maintain situational awareness and proper spacing from other aircraft in the traffic pattern.

Forward Slip to Landing

- Student says, student does.
 - Verify correct procedure.
 - Power must be at idle.
 - Ailerons into the wind.
 - Rudder opposite.
- Standards:
 - Configure the airplane correctly.
 - As necessary, correlate crosswind with direction of forward slip and transition to sideslip before touchdown.
 - Touch down at a proper pitch attitude, within 400 feet beyond or on the specified point, with no side drift, and with the airplane's longitudinal axis aligned with and over the runway center/landing path.
 - Maintain a ground track aligned with the runway center/landing path.

Go-around or rejected landings

- If the student has an unstable approach, see if they decide to initiate a go-around.
 - If all of the approaches are stable, have the student perform a go-around due to "something" on the runway (airplane, animal, vehicle, etc.)
 - Verify correct procedures.
- Standards:
 - Apply takeoff power immediately and transition to climb pitch attitude for VX or VY as appropriate $+10/-5$ knots.
 - Configure the airplane after a positive rate of climb has been verified or in accordance with airplane manufacturer's instructions.
 - Maintain VY $+10/-5$ knots to a safe maneuvering altitude.
 - Maintain directional control and proper wind-drift correction throughout the climb.

Normal and crosswind landing

- Student says, student does.
 - Verify correct procedure.
 - Ensure proper crosswind correction.
- Standards:
 - Select and aim for a suitable touchdown point considering the wind, landing surface, and obstructions.
 - Establish the recommended approach and landing configuration and airspeed and adjust pitch attitude and power as required to maintain a stabilized approach.
 - Maintain directional control and appropriate crosswind correction throughout the approach and landing.
 - Touch down at a proper pitch attitude, within 400 feet beyond or on the specified point, with no side drift, and with the airplane's longitudinal axis aligned with and over the runway center/landing path.

Short-field approach and landing

- Student does, student says:
 - Verify correct procedure.
 - Emphasize the importance of not forcing the airplane on the ground. If the point cannot be made, perform a go-around.
 - Verify simulated maximum braking.
- Standards:
 - Maintain manufacturer's published approach airspeed or in its absence not more than 1.3 VSO, +10/-5 knots with gust factor applied.
 - Touch down at a proper pitch attitude within 200 feet beyond or on the specified point.

Soft-field approach and landing

- Student does, student says:
 - Verify correct procedure.
 - Verify minimal braking.
- Standards:
 - Maintain manufacturer's published approach airspeed or in its absence not more than 1.3 VSO, +10/-5 knots with gust factor applied.
 - Make smooth, timely, and correct control inputs during the round out and touchdown, and, for tricycle gear airplanes, keep the nose wheel off the surface until loss of elevator effectiveness.
 - Touch down at a proper pitch attitude with minimum sink rate, no side drift, and with the airplane's longitudinal axis aligned with the center of the runway.
 - Maintain elevator as recommended by manufacturer during rollout and exit the "soft" area at a speed that would preclude sinking into the surface.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 149a.
- Assign homework:
 - Review for the EOC mock.
 - Plan one of the TOLD card/XC problems from the Florida Tech Aviation HUB.

Completion Standards

This lesson is complete when:

1. The applicant demonstrates proficiency for all line items that meets or exceeds the standard of performance outlined in the current FAA Private Pilot ACS.